

olmoearth_inferenceX: what was done, what held, and what did not

Technical report, not a submission

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https://github.com/2imi9/olmoearth_inferenceX

<https://olmoearth-inferencex.readthedocs.io>

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Summary

A prediction map from an Earth-observation foundation model ships over regions that have no labels. This project asks one question about such a map, with no labels at hand: *which windows should a reviewer check first, why is each one suspect, and when two inferences of the same scene disagree, how much do they differ, where, and can the map itself say which is right?* Labels are used only to grade the answers, never to produce them.

The answer, across 65 experiments and 146 recorded claims, is short. The model’s own confidence, the top-1 minus top-2 logit margin on each window, is the signal. It ranks the model’s own errors on every expert-labelled testbed tried, on five kinds of reference that fail in different ways, and on all 24 tasks of the model authors’ own published evaluation suite, which this project did not choose, under OlmoEarth and fifteen other encoders. Nothing else tried beats it, on the water testbeds or on that suite: not ensembles, not distance from the training distribution, not consistency under perturbation, not the pretraining objective’s own error, not a second model of the same family. Comparing two inferences of one scene says how much they differ and where, and the answer to *which side is right* is, without labels, to decline: the more confident side is right on only half to two thirds of the differing windows.

Three of the project’s own predictions failed and are kept. A strong reasoning model given a numpy sandbox rediscovers the ranking by itself and grounds its numbers nearly as well as a model given the package as tools; what the package adds to such a model is the evidence about when not to choose a side, and reliability; at 7B the same package is decisive on every axis, and the agent no longer finds it unaided. The project’s method is its other contribution: predictions written before each run with what would falsify them, every documented number pinned to a committed artifact by an executable check, and 31 rejected claims recorded as prominently as the 40 supported ones.

This report is written so that Ai2 can read what was done in an hour and decide whether any of it is worth refining together. It is not a paper and does not claim novelty; each idea’s origin is in Section 8.

1 The question, and what is not asked

Given a map, three questions. Which windows to review first at a fixed budget, say 5% of the windows. Why each flagged window is suspect, in terms a reviewer can check. And, given a second inference of the same windows (another crop offset, another backbone, another sensor,

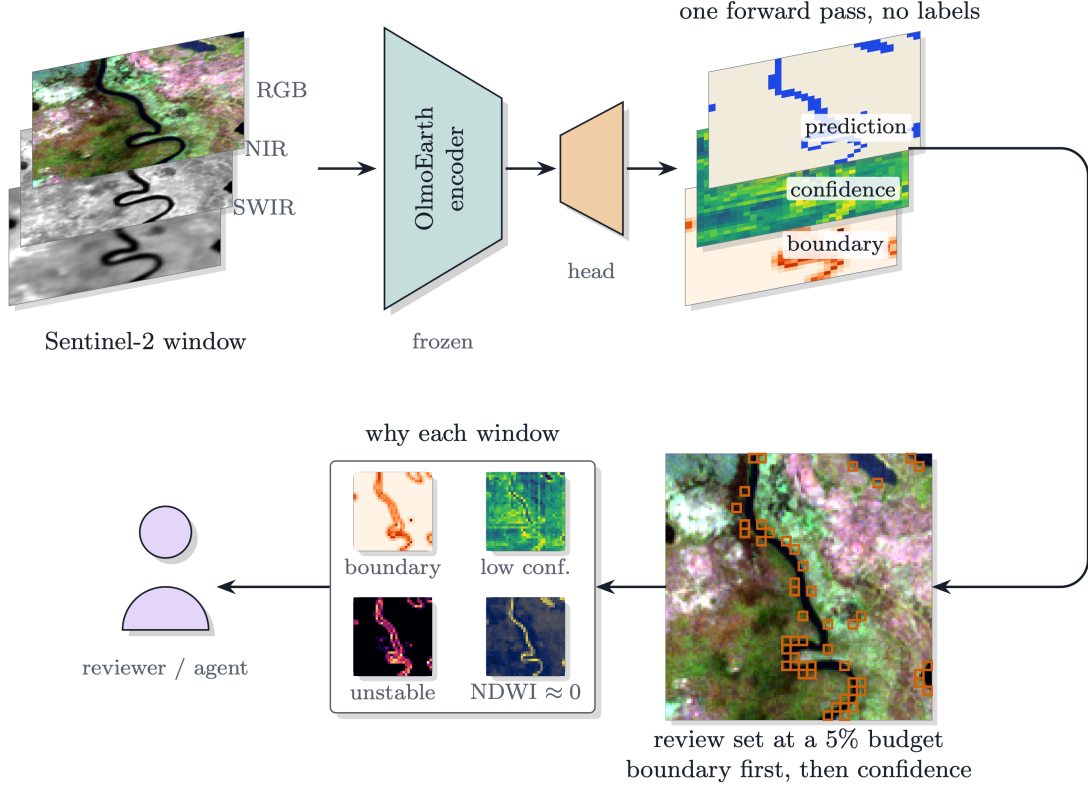


Figure 1: One scene through the audit: Sentinel-2 bands, the frozen OlmoEarth encoder and a task head, the prediction with its confidence and boundary layers, the review set at a 5% budget, and the reasons per flagged window. One forward pass, no labels.

another date, before against after fine-tuning), how much the two differ, where the differences sit, and which to believe.

What is not asked: whether the model is accurate. Accuracy is what the model’s own benchmarks measure. This measures where a given map is wrong, which no public benchmark scores. It is also not out-of-distribution detection: a window can be far outside the training distribution and right, or inside it and wrong, and the signal here tracks the second.

2 Protocol

2.1 Units, signal, statistics

The graded unit is a 4-pixel window on the prediction grid, or the sample on classification tasks. The signal is the model’s own margin, top-1 minus top-2 logit; its negation is the suspicion score, higher first. Ranking quality is the excess area under the risk-coverage curve [E-AURC, Geifman et al., 2019], tie-aware, and the capture at a budget: the share of all errors inside the most suspect 5%, 10% or 20% of windows. A budget of b on a map with error rate e cannot capture more than $\min(1, b/e)$, so every capture is reported beside that attainable ceiling. Comparisons between signals use one-sided exact sign tests over independent groups (tiles, events, regions, tasks) and, where a reference is a stratified sample, design-weighted estimators [Olofsson et al., 2014] with a cluster bootstrap.

2.2 Two baselines every rule must beat

A candidate signal is scored against the model’s own confidence and against a control that sees no model: a pixel statistic such as a water index or the within-window variance, or, where imagery is not published, the distance of a sample from the training set’s mean embedding and the rarity of the predicted class. A signal that does not beat the no-model control does not exist; one that beats it and loses to confidence is the interesting case, and most of Section 4 is that case.

2.3 Five kinds of reference

The answer key came from	Testbeds
people reading the same imagery	Sen1Floods11 hand labels [Bonafilia et al., 2020], Copernicus EMS through GEOID-Flood [Chiriaco et al., 2026], World-Floods v2
another model’s output	DFC2020 [Yokoya et al., 2020], whose test labels are an iterated random forest
experts annotating a served product	Dynamic World [Brown et al., 2022], 409 expert tiles, audited from its own published probabilities
surveyors standing in the field	LUCAS Copernicus 2022 [d’Andrimont et al., 2021], the only reference here that never saw a pixel
farmers’ own declarations	EuroCrops [Schneider et al., 2023], which also labels both sides of a two-date comparison

Table 1: A signal that survives five kinds of answer key is unlikely to be an artifact of one.

2.4 Preregistration and the claim ledger

Every experiment states its predictions and their falsification conditions in its docstring before it runs; a null is not equivalence. Every sentence in the documentation that states a number carries a claim id, and `docs/claims.yaml` holds 138 claims, each with a statement, a status, the experiments and artifacts it rests on, and an executable check; `scripts/claims.py` `stale` fails when an artifact no longer says what the prose says. The statuses at the time of writing: 40 supported, 44 measured, 21 mixed, 31 rejected, 2 superseded. Labels grade signals and never train them [labels-grade-never-train].

3 What holds

3.1 Confidence ranks the errors

The model’s own confidence is the best single label-free ranker on the AWF points, the fine-tuned model and the Sen1Floods11 test split, under both backbones and both window designs, leading or tying [confidence-best-single-signal]. In a reviewer’s units, on the median GEOID-Flood area of interest the 5% of windows it flags hold 88% of the permanent-water errors and 64% of the post-event water errors, 17.5 and 12.8 times a random 5% [geoid-capture-effect-size].

The exception is recorded. On one flood event, Bolivia, a plain water index matches confidence, because water against not-water there is close to a spectral threshold; over 45 GEOID areas from nine activations the honest exception rate is one activation in nine [geoid-exception-rate], and the exception belongs to the frozen encoder: after fine-tuning with the model authors’ recipe it is gone [finetune-dissolves-bolivia-exception].

3.2 Tasks this project did not choose

The strongest objection to testbeds one picked oneself is answered by the model authors’ own published embedding suite: 25 tasks chosen by them for their paper [Herzog et al., 2025], splits fixed in the files, no encoder pass needed. On all 24 tasks where a margin is defined, and on all 14 distinct sources behind them, the margin beats the best no-model control: 6,435,473 graded units, an accuracy range from 0.333 to 0.979, 17 of 17 classification and 7 of 7 segmentation tasks, sign test $p = 6 \times 10^{-8}$; six of the tasks come from GEO-Bench [Lacoste et al., 2023] and the margin wins on all six [suite-margin-wins-every-task].

The bar a practitioner cares about is the alternatives the literature actually proposes, run on the same tasks (exp73): a five-seed probe ensemble [Lakshminarayanan et al., 2017], distance to the tenth nearest training embedding [Sun et al., 2022], and a class-conditional Mahalanobis distance [Lee et al., 2018] fitted on predicted classes so that labels never train it. Preregistered at exp70’s own bar, all three hold: the margin beats the ensemble on 22 of 24 tasks ($p = 1.8e - 05$), the nearest-neighbour distance on 24 of 24 and the Mahalanobis distance on 24 of 24 ($p = 6.0e - 08$) [suite-margin-beats-the-strong-alternatives]. The alternatives are real signals, each beating the no-model control on roughly half the suite (15, 14 and 12 tasks), and better than the margin on none; the ensemble is nearly the margin’s equal on classification and clearly behind on segmentation [suite-alternatives-clear-the-control-and-still-lose].

One more input was tried after the release. The same tasks are published embedded from different sensors, so a head fitted on another sensor is an expert on a different input, and its disagreement with the reference head costs no labels (exp75, five groups, 464,366 units). That disagreement beats every no-model control [sensor-disagreement-beats-the-control], and still does not pay: of the errors outside the margin’s 10% review set it finds more than the control on three of five groups and not on the largest, and interleaving the two rankings captures no more at 5% or 10% on any group [sensor-disagreement-does-not-pay-in-a-review-set]. What the margin misses is mostly invisible to a second sensor as well; closing it needs labels.

Which member of the confidence family is the last question asked of the suite (exp76). Under AUGRC, which for fixed errors orders readings as the failure AUROC does, the margin beats the best control on 23 of 24 tasks [augrc-leaves-the-suite-result-standing]. One minus the top probability ranks errors better than top-1 minus top-2 on 14 of the 16 multi-class tasks, by a median 0.0006 of excess AURC beside a lead over the control of 0.116, and on binary tasks the forms are one ranking [top1-beats-the-margin-on-multiclass]; no score built from the whole logit vector, and no aggregation of pixels into a window, does better.

Nor is the result a property of one encoder. Ai2 published the same task embeddings for other models with their own probe settings, so exp70’s protocol was run unchanged on every encoder carrying at least 20 of the 24 tasks (exp74): fifteen encoders, from eight families outside OlmoEarth (AnySat, Clay, Panopticon, Galileo, CROMA, TerraMind, Satlas, Copernicus-FM) and the OlmoEarth size series. Across 332 scored (encoder, task) pairs and 92 million graded units the margin beats the best no-model control on 322, every encoder at 90% or better (preregistered P1, bar 75%), eight of them undefeated; the ten losses sit on CropHarvest Togo and Nandi Sentinel-1 alone [suite-holds-under-every-encoder]. The three outside families that carry the whole suite win on 23, 24 and 24 of 24 (P2, bar 22) [suite-outside-families-hold-at-twenty-two], and across the OlmoEarth sizes the median lead is flat to slightly narrowing, +0.131 at tiny to +0.113 at large, while the tasks won rise from 22 to 24 (P3, descriptive) [suite-lead-across-olmoearth-sizes]. A task an encoder was not published for is recorded as absent, never as a loss.

3.3 Products this project had no hand in

Dynamic World can be audited from nothing but the probabilities it publishes about itself. On its 409 expert-annotated tiles, 4,348,526 windows, the product’s own margin ranks its own errors

better than a control that sees neither the imagery nor the probabilities: pooled E-AURC 0.0637 against 0.1120, 315 tiles to 91, $p = 2.7 \times 10^{-30}$; as a check, the pooled decision reproduces the product’s published top-1 on 98.99% of windows [dw-margin-ranks-a-production-model]. The margin also beats one minus the top-1 probability there, and the reason is ties: the naive form takes the same value on 23.4% of windows, the margin on 3.6% [dw-margin-beats-naive-confidence]. The explanation cues transfer: among the product’s error windows the boundary cue fires 2.93 times as often as among its correct ones [dw-errors-carry-the-cues].

On LUCAS, where a surveyor stood at the point and the reference never saw a pixel, the margin’s design-weighted E-AURC is 0.1468 against 0.2421 for the best control an operator could compute at inference time, on 94 regions to 32 [lucas-ranking-survives-ground-observation]. On EuroCrops, farmers’ declarations across three countries, it beats the best control in every region and on all but four of 146 grid cells [eurocrops-ranking-holds-on-declarations]. On DFC2020’s eight classes it beats the pixel control on all three sensor arms [dfc2020-margin-beats-pixel-control].

3.4 Comparing two inferences of one scene

The package’s second command measures the difference between two inferences on identical windows. Across every pair tried, crop offsets, backbones, sensors, encoders, frozen against fine-tuned, the differing windows sit on class boundaries, enriched 3.3 to 7.1 times on Sen1Floods11 and 21 times on the median GEOID event [atlas-disagreement-is-boundary-located]; the sensor difference and the backbone difference are different sets of windows, ϕ 0.27 and 0.18 [atlas-different-differences]. A same-period cross-sensor difference carries none of a later flood, 0.4% of its windows against 19.5% for a mixed pair [two-periods-sensor-axis-isolated]; on DFC2020 the two sensors differ on 44.27% of windows, 33 times what reseeding the probe moves, with the optical side right on 50.4%, the radar side on 17.2% and both wrong on 32.4% [dfc2020-sensor-difference-dominates-land-cover].

Which side is right stays a labelled question. Two facts fix that reading. The raw margin rule, believe the more confident side, is right on 55 to 61% of differing windows; a rule fitted on labels reaches 73 to 75% but only within the model family it was fitted on, and forced across families it falls to 44 to 49% [calibrate-family-lock]. So the tool declines, and says why.

EuroCrops labels both years of a two-date comparison and gives the floor such a comparison needs: the map moves on 6% to 37% of windows whose declared crop did not change, one to two orders of magnitude above the 0.3% that reseeding the readout moves [eurocrops-the-labelled-floor-for-a-two-date-difference]; and on 5% to 37% of differing windows the model was right about both years, which is the model correctly following a rotation and not either side being wrong [eurocrops-a-difference-can-be-the-model-tracking-the-ground].

3.5 The review order, the explanation layer, and the labelled fusion

Reviewing boundary windows first, then the interior in confidence order, captures more of the errors than confidence alone at the 5% and 10% budgets and is no longer ahead at 20% [boundary-first-review-order]; on a served product that exports no confidence, the boundary fraction alone captures a median 0.88 of the water disagreements at 5% [served-product-boundary-triage]. With labels where they exist, a fusion of five label-free readings, cross-fitted by tile, cuts confidence’s E-AURC from 0.0105 to 0.0084 on Bolivia and from 0.0098 to 0.0069 on the test split [calibrate-ranker-fusion-beats-confidence], bound to the family it was fitted on.

4 What did not work

Thirty-one claims are rejected. Grouped, with the reason, because the reason is what a reader can reuse:

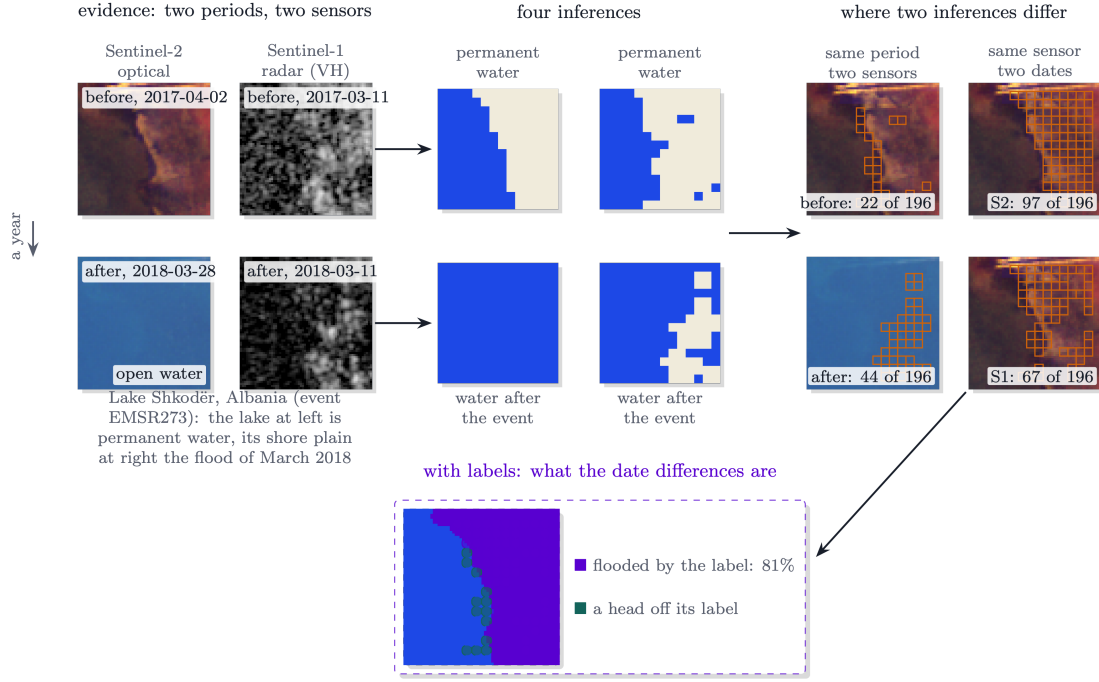


Figure 2: Four dated inputs of one GEOID-Flood chip, each read on identical windows; the same-period and same-sensor differences on the scene, and the label bridge boxed apart, because what a difference *is* needs labels.

None of this is surprising to the failure-detection literature, where plain confidence has been the baseline most proposed methods fail to beat [Hendrycks and Gimpel, 2017, Jaeger et al., 2023]. What is measured here is that the same holds on Earth-observation maps at the window level, against five kinds of reference.

5 Does the package help an agent?

Everything above is about the measurement. Whether the tool helps was preregistered and run as a benchmark (exp64): forty cards from the labelled testbeds with hidden labels and a 5% budget, ten with a second inference, three tasks (the review set, the reasons, the comparison with declining as the correct answer on which side), four arms sharing one model, and a claims audit that matches every number an arm states against that run’s tool outputs. The model is Qwen3.8-27B in NVFP4, served by vLLM; the plan asked for 30 to 70 billion parameters in fp16 or int8, and both deviations are recorded. A fifth arm the plan left open, the OlmoEarth Agent as it ships with the package’s skill among its forty-one tools, was run and is reported descriptively.

Handed the package as tools, the model reproduces the package; the agent as shipped chose the package’s review-set tool on 40 of 40 runs and its compare tool on 10 of 10 comparison cards without being pointed at either, and did the same [agent-benchmark-tool-arm-reproduces-the-package]. Two of the three predictions failed. The sandbox arm captured 0.904 of the package’s errors by itself, which is the falsification the plan named in advance [agent-benchmark-sandbox-rediscovers-the-ranking]; and it grounded 94.2% of its numbers against the tool arm’s 99.4%, below the preregistered gap. The first grading of that second prediction had said it held: the audit had read numbers from JSON fields and skipped strings, and a sandbox’s only tool output is what it prints. That reading was withdrawn [agent-benchmark-grounding-not-decisive]. The prediction that held is the decline: the tool arm declines the side question on every comparison card, the sandbox on one in ten, because the fact that confidence does not settle which side is right is carried by the package and not derivable

Family	What was measured, and why it loses
Ensembles and posteriors over the head	Bootstrap bags, probe-seed ensembles, mutual information, a Laplace posterior. Members of one family err on the same windows; the disagreement adds noise to the margin, not information. [internal-state-signals-rejected]
Feature-space typicality	kNN, Mahalanobis, embedding distance, feature-space neighbourhood contradiction. They measure whether the model is outside what it knows, which is a different question from whether this window is wrong. [embedding-dissimilarity-rejected]
A second model	Another backbone version, another encoder family, an outside representation. Every encoder tried errs on 80% of OlmoEarth’s error windows (ϕ 0.77–0.81), so two-view disagreement captures 0.24 of the errors at 10% against 0.45 for confidence. [two-view-disagreement-rejected]
Consistency under perturbation	Flip-and-rotate, masking, test-time augmentation over crop offsets. Instability tracks context reliance and texture, not error. [dihedral-consistency-rejected]
The pretraining objective	The masked-token decoder’s own residual, with the native, a whitened and a discrete target. The frozen target has effective rank 2 on real scenes; a reconstruction objective tracks the variance subspace, not the task [Esser et al., 2025, Balestrieri and LeCun, 2024]. [retargeted-residual-rejected]
Published reliability signals	SHRUG-FM’s [Gonzalez-Calabuig et al., 2025] image-level signals ported to the window: confidence beats each of them, and its own paper never ran the single-model confidence baseline. [shrug-signals-rejected]
Label-free reliability estimation	Dawid–Skene over a model family inflates every member and inverts the ordering, because the family errs together. [family-errs-together-dawid-skene]

Arm	numbers grounded	capture, share of package’s	cues that hold	declines the side question	runs with no answer
A, the package as tools	99.4%	1.000	99.5%	10/10	0/120
B, numpy sandbox, no package	94.2%	0.904	86.2%	1/10	27/120
C, template, no model	100%	1.000	100%	10/10	0/40
D, description only, no rasters	–	0.228	23.6%	10/10	0/120
E, the OlmoEarth Agent, as shipped	99.8%	0.963	98.7%	10/10	1/40

Table 2: exp64 on Qwen3.8-27B, forty cards. No arm fabricated a window.

from two arrays [agent-benchmark-decline-holds]. Reliability separates the arms where ranking does not: 27 of the sandbox’s 120 runs ended without an answer [agent-benchmark-sandbox-unreliable].

The same forty cards at 7B (Qwen2.5-7B-Instruct, preregistered after the 27B result) measure the other end: all three predictions hold there, grounding 100.00.144 of the package’s, declines 10 of 10 against 0, with the sandbox failing to answer in 51 of 120 runs [agent-benchmark-package-is-a-floor-at-7b]. And the agent at 7B does not find the tool on its own, 17 of 40 runs against 40 of 40 at 27B, which rejects the prediction that it would reproduce the package at that size; pinning the skill recovers most of it [agent-benchmark-7b-agent-does-not-find-the-tool].

The reading, stated plainly: for a model of this strength the ranking is not what the package adds; for a weaker one it is, on every axis. What it adds is calibrated knowledge, when computation does not settle the question, and a reliable path to the same answer. The cards are binary water tasks and the sandbox was handed the margin by name, so the harder test, multi-class cards with raw per-class scores where the model must choose a signal, is open.

6 Two warnings for practitioners

Report the readout’s generalisation gap beside any ranking comparison. A small probe that has memorised its practice data reverses which confidence signal ranks best; on LUCAS the margin went from last of four model signals to the front, on the same data, purely from choosing regularisation on held-out ground. Across the 24-task suite the ordering degrades with the gap by four ten-thousandths rather than inverting, so only severe memorisation flips it [suite-probe-gap-degrades-the-ordering]. The common practice of filtering a land-cover reference to large homogeneous units flatters the audit rather than understating it: AUROC 0.774 on the units the filter keeps against 0.677 on the ones it deletes [lucas-the-homogeneity-filter-flatters-the-tool].

A two-date difference needs the right floor. Comparing a date difference against the rate at which reseeded the readout moves the map flatters it by one to two orders of magnitude; the floor has to be measured on a reference that labels both dates.

7 Limits

Most testbeds are binary water tasks, on which the margin, one minus top-1 and entropy coincide; the multi-class evidence is DFC2020, Dynamic World, LUCAS, EuroCrops and the 24-task suite. The encoders are the OlmoEarth family plus one served product; the competitors in Section 4 were beaten on the water testbeds and, for three of the families, on the 24-task suite as well. One author; nothing here has been reproduced by anyone else, which is a research criterion this project cannot meet alone. Several experiments assume the cluster’s paths. The agent benchmark used one model below the plan’s precision. The package is two commands and a small API, deliberately.

8 Where each idea comes from

Nothing here is claimed as new in isolation. The reject option and risk-coverage curves are selective classification [Chow, 1970, Geifman and El-Yaniv, 2017]; E-AURC is Geifman et al. [2019]; confidence as the baseline is Hendrycks and Gimpel [2017], and that most alternatives fail to beat it is Jaeger et al. [2023]; the margin rule is margin sampling [Scheffer et al., 2001] in the uncertainty-sampling tradition [Lewis and Gale, 1994]; segment-level error prediction from softmax dispersion is Rottmann et al. [2020]; distance-to-training warnings in spatial models are the area of applicability [Meyer and Pebesma, 2021]; design-weighted map accuracy is Olofsson et al. [2014]; multi-scale EO models the package was compared against include Galileo [Tseng et al., 2025]; and grading an agent’s grounding separately from its accuracy follows Bansal et al. [2021]. What this project did is run the known signals head to head, at the window level, on Earth-observation maps, against five kinds of reference and the model authors’ own task suite, under a protocol that records what fails.

9 What this is for, and what could be refined together

The package answers three questions about a map with no labels and carries the evidence for its answers into its outputs, which is what lets an agent state them and decline the fourth. Concretely for Ai2: the served products’ own probabilities are enough to audit them, as Dynamic World shows; the agent skill is built, cross-checked bit-exact against the package on real inference, and merged into the OlmoEarth Agent repository’s main branch; and the protocol scores any new candidate rule against the same two baselines on the same references.

Open items, in the order they would help: the multi-class agent benchmark; a fine-tuned land-cover head exported from the platform, to audit as Dynamic World was; and an independent

reproduction of any one result by someone who is not the author.

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A The package, briefly

`oe-inferencex assess` takes a probability or logit raster and writes the review set at each budget with pixel and map coordinates, a suspicion raster, a boundary raster and an explanation file. `oe-inferencex compare` takes two maps on one grid and writes the disagreement rate, the differing windows with their cues, and, with labels, which side was right. `oe_inferencex.metrics` holds the tie-aware E-AURC, the capture at a budget, the attainable ceiling and their design-weighted forms; `oe_inferencex.stats` the sign tests and cluster bootstraps; `oe_inferencex.calibrate` the label-fitted fusion with its family lock. The package needs no torch; the experiments need the encoder.

B Where the numbers live

Every number in this report is in `docs/` with a claim marker: `Findings.md` for what holds and what was rejected, `TECHNIQUES.md` for everything tried in one line each, `results/comparisons.md` for the per-experiment sections, `claims.yaml` for the checks, `related_work.md` for the origins, and `exp/out/` for the artifacts the checks read.